Software Engineering Lab - Lab 6

Code and Branch Coverage Test Suite & Terminal Output

Course:	UE24CS341A - Software Engineering Lab	**Date:**	November 03, 2026
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Department:	Computer Science & Engineering, PES University	**Module Under Test:**	food_delivery.py

1. Code Coverage Test File: CodeCoverageTest_food_delivery.py

```
import pytest
from food_delivery import delivery_charge, apply_coupon, estimated_time

# ============================================================================
# CODE COVERAGE TEST SUITE
# Designed to achieve 100% Line (Statement) Coverage on food_delivery.py
# ============================================================================

# ----------------------------------------------------------------------------
# Tests for delivery_charge(distance_km, order_value, is_raining)
# ----------------------------------------------------------------------------

def test_delivery_charge_out_of_range():
    """Executes lines 5->6: raises ValueError when distance is invalid."""
    with pytest.raises(ValueError, match="out of delivery range"):
        delivery_charge(0, 200, False)

def test_delivery_charge_free_delivery_over_500():
    """Executes lines 7->8: charge = 0 when order_value >= 500."""
    assert delivery_charge(5, 500, False) == 0

def test_delivery_charge_distance_upto_3km():
    """Executes lines 9->10: charge = 20 for short distance (<= 3 km)."""
    assert delivery_charge(2, 200, False) == 20

def test_delivery_charge_distance_upto_8km():
    """Executes lines 11->12: charge = 40 for medium distance (<= 8 km)."""
    assert delivery_charge(6, 200, False) == 40

def test_delivery_charge_distance_above_8km():
    """Executes lines 13->14: charge = 60 for long distance (> 8 km)."""
    assert delivery_charge(10, 200, False) == 60

def test_delivery_charge_with_rain_surcharge():
    """Executes lines 15->16: adds 30 surcharge when is_raining is True."""
    assert delivery_charge(2, 200, True) == 50


# ----------------------------------------------------------------------------
# Tests for apply_coupon(order_value, code, is_first_order)
# ----------------------------------------------------------------------------

def test_apply_coupon_none():
    """Executes lines 21->22: returns unchanged order_value when code is None."""
    assert apply_coupon(300, None, False) == 300

def test_apply_coupon_invalid_code():
    """Executes lines 23->24: raises ValueError when coupon code is not recognized."""
    with pytest.raises(ValueError, match="invalid coupon"):
        apply_coupon(300, "DISCOUNT50", False)

def test_apply_coupon_welcome_not_first_order():
    """Executes lines 25->26: raises ValueError when WELCOME coupon used on non-first order."""
    with pytest.raises(ValueError, match="coupon only for first order"):
        apply_coupon(300, "WELCOME", False)

def test_apply_coupon_order_below_minimum_value():
    """Executes lines 27->28: returns original order_value if under 200."""
    assert apply_coupon(150, "SAVE50", False) == 150

def test_apply_coupon_success():
    """Executes line 29: deducts coupon discount from eligible order."""
    assert apply_coupon(350, "SAVE50", False) == 300


# ----------------------------------------------------------------------------
# Tests for estimated_time(distance_km, is_peak_hour, restaurant_busy)
# ----------------------------------------------------------------------------

def test_estimated_time_peak_hour_and_busy():
    """Executes lines 33, 34->35, 36->37, 38->39, 40:
    All conditional branches evaluate to True, executing every line in function.
    Base (15 + 5*4 = 35) + peak (10) + busy (15) + peak-busy (5) = 65 mins.
    """
    assert estimated_time(5, True, True) == 65
```

Code Coverage Terminal Output Screenshot (100% Line Coverage)

```
PS C:\Users\Lenovo\Desktop\Sem-5\SE\SE-lab\lab-6\L6_L7_20_food_delivery\Lab 6> coverage run -m pytest CodeCoverageTest_food_delivery.py
============================= test session starts =============================
platform win32 -- Python 3.12.10, pytest-9.2.0, pluggy-1.7.0
rootdir: C:\Users\Lenovo\Desktop\Sem-5\SE\SE-lab\lab-6\L6_L7_20_food_delivery\Lab 6
configfile: pytest.ini
plugins: anyio-4.15.1, platformdirs-4.12.2
collected 12 items

CodeCoverageTest_food_delivery.py ............                           [100%]

============================== 12 passed in 0.05s ==============================

PS C:\Users\Lenovo\Desktop\Sem-5\SE\SE-lab\lab-6\L6_L7_20_food_delivery\Lab 6> coverage report -m
Name                               Stmts   Miss  Cover   Missing
----------------------------------------------------------------
CodeCoverageTest_food_delivery.py     29      0   100%
food_delivery.py                      33      0   100%
----------------------------------------------------------------
TOTAL                                 62      0   100%

PS C:\Users\Lenovo\Desktop\Sem-5\SE\SE-lab\lab-6\L6_L7_20_food_delivery\Lab 6> coverage run --branch -m pytest CodeCoverageTest_food_delivery.py
============================= test session starts =============================
collected 12 items
CodeCoverageTest_food_delivery.py ............                           [100%]
============================== 12 passed in 0.05s ==============================

PS C:\Users\Lenovo\Desktop\Sem-5\SE\SE-lab\lab-6\L6_L7_20_food_delivery\Lab 6> coverage report -m
Name                               Stmts   Miss Branch BrPart  Cover   Missing
-------------------------------------------------------------------------------
CodeCoverageTest_food_delivery.py     29      0      0      0   100%
food_delivery.py                      33      0     24      3    95%   34->36, 36->40, 38->40
-------------------------------------------------------------------------------
TOTAL                                 62      0     24      3    97%

PS C:\Users\Lenovo\Desktop\Sem-5\SE\SE-lab\lab-6\L6_L7_20_food_delivery\Lab 6>
```

2. Branch Coverage Test File: BranchCoverageTest_food_delivery.py (Part 1)

```
import pytest
from food_delivery import delivery_charge, apply_coupon, estimated_time

# ============================================================================
# BRANCH COVERAGE TEST SUITE
# Designed to achieve 100% Branch Coverage on food_delivery.py
# Tests every decision branch (True and False), boundary conditions, and edge cases.
# ============================================================================

# ----------------------------------------------------------------------------
# 1. Tests for delivery_charge(distance_km, order_value, is_raining)
# ----------------------------------------------------------------------------

def test_delivery_charge_invalid_distance_lower_bound():
    """Branch: distance_km <= 0 (True via distance <= 0)."""
    with pytest.raises(ValueError, match="out of delivery range"):
        delivery_charge(0, 200, False)

def test_delivery_charge_invalid_distance_negative():
    """Branch: distance_km <= 0 (True via negative distance)."""
    with pytest.raises(ValueError, match="out of delivery range"):
        delivery_charge(-2, 200, False)

def test_delivery_charge_invalid_distance_upper_bound():
    """Branch: distance_km > 15 (True via distance > 15)."""
    with pytest.raises(ValueError, match="out of delivery range"):
        delivery_charge(15.1, 200, False)

def test_delivery_charge_free_delivery_no_rain():
    """Branch: order_value >= 500 (True) with is_raining (False)."""
    assert delivery_charge(5, 500, False) == 0

def test_delivery_charge_free_delivery_with_rain():
    """Branch: order_value >= 500 (True) with is_raining (True surcharge)."""
    assert delivery_charge(5, 600, True) == 30

def test_delivery_charge_tier1_boundary_no_rain():
    """Branch: distance_km <= 3 (True boundary 3km), is_raining (False)."""
    assert delivery_charge(3, 499, False) == 20

def test_delivery_charge_tier1_with_rain():
    """Branch: distance_km <= 3 (True), is_raining (True)."""
    assert delivery_charge(1, 200, True) == 50

def test_delivery_charge_tier2_boundary_no_rain():
    """Branch: distance_km <= 8 (True boundary 8km), is_raining (False)."""
    assert delivery_charge(8, 200, False) == 40

def test_delivery_charge_tier2_with_rain():
    """Branch: distance_km <= 8 (True), is_raining (True)."""
    assert delivery_charge(5, 200, True) == 70

def test_delivery_charge_tier3_max_distance_no_rain():
    """Branch: distance_km <= 8 (False, boundary 15km), is_raining (False)."""
    assert delivery_charge(15, 200, False) == 60

def test_delivery_charge_tier3_with_rain():
    """Branch: distance_km > 8 (True), is_raining (True)."""
    assert delivery_charge(10, 200, True) == 90


# ----------------------------------------------------------------------------
```

Branch Coverage Test File: BranchCoverageTest_food_delivery.py (Part 2)

```
# 2. Tests for apply_coupon(order_value, code, is_first_order)
# ------------------------------------------------------------------------------

def test_apply_coupon_none():
    """Branch: code is None (True)."""
    assert apply_coupon(300, None, False) == 300

def test_apply_coupon_not_in_coupons():
    """Branch: code not in COUPONS (True)."""
    with pytest.raises(ValueError, match="invalid coupon"):
        apply_coupon(300, "UNKNOWN_CODE", True)

def test_apply_coupon_welcome_not_first_order():
    """Branch: code == 'WELCOME' and not is_first_order (True)."""
    with pytest.raises(ValueError, match="coupon only for first order"):
        apply_coupon(300, "WELCOME", False)

def test_apply_coupon_welcome_first_order_eligible():
    """Branch: code == 'WELCOME' and is_first_order is True, order >= 200."""
    assert apply_coupon(300, "WELCOME", True) == 200

def test_apply_coupon_welcome_first_order_ineligible():
    """Branch: code == 'WELCOME' and is_first_order is True, order < 200."""
    assert apply_coupon(150, "WELCOME", True) == 150

def test_apply_coupon_save50_order_below_200():
    """Branch: code == 'SAVE50', order_value < 200 (True)."""
    assert apply_coupon(199, "SAVE50", False) == 199

def test_apply_coupon_save50_boundary_200():
    """Branch: code == 'SAVE50', order_value >= 200 (boundary 200)."""
    assert apply_coupon(200, "SAVE50", False) == 150

def test_apply_coupon_save50_order_above_200():
    """Branch: code == 'SAVE50', order_value >= 200 (above minimum)."""
    assert apply_coupon(400, "SAVE50", True) == 350


# ------------------------------------------------------------------------------
# 3. Tests for estimated_time(distance_km, is_peak_hour, restaurant_busy)
# ------------------------------------------------------------------------------

def test_estimated_time_normal_conditions():
    """Branches: is_peak_hour (False), restaurant_busy (False).
    minutes = 15 + 5*4 = 35.
    """
    assert estimated_time(5, False, False) == 35

def test_estimated_time_peak_hour_only():
    """Branches: is_peak_hour (True), restaurant_busy (False).
    minutes = 15 + 5*4 + 10 = 45.
    """
    assert estimated_time(5, True, False) == 45

def test_estimated_time_restaurant_busy_only():
    """Branches: is_peak_hour (False), restaurant_busy (True), inner is_peak_hour (False).
    minutes = 15 + 5*4 + 15 = 50.
    """
    assert estimated_time(5, False, True) == 50

def test_estimated_time_peak_hour_and_restaurant_busy():
    """Branches: is_peak_hour (True), restaurant_busy (True), inner is_peak_hour (True).
    minutes = 15 + 5*4 + 10 + 15 + 5 = 65.
    """
    assert estimated_time(5, True, True) == 65
```

Branch Coverage Terminal Output Screenshot (100% Branch Coverage)

```
PS C:\Users\Lenovo\Desktop\Sem-5\SE\SE-lab\lab-6\L6_L7_20_food_delivery\Lab 6> coverage run --branch -m pytest BranchCoverageTest_food_delivery.py
============================= test session starts =============================
platform win32 -- Python 3.12.10, pytest-9.2.0, pluggy-1.7.0
rootdir: C:\Users\Lenovo\Desktop\Sem-5\SE\SE-lab\lab-6\L6_L7_20_food_delivery\Lab 6
configfile: pytest.ini
plugins: anyio-4.15.1, platformdirs-4.12.2
collected 23 items

BranchCoverageTest_food_delivery.py .......................              [100%]

============================== 23 passed in 0.08s ==============================

PS C:\Users\Lenovo\Desktop\Sem-5\SE\SE-lab\lab-6\L6_L7_20_food_delivery\Lab 6> coverage report -m
Name                                Stmts   Miss Branch BrPart  Cover   Missing
-------------------------------------------------------------------------------
BranchCoverageTest_food_delivery.py      53      0      0      0   100%
food_delivery.py                         33      0     24      0   100%
-------------------------------------------------------------------------------
TOTAL                                    86      0     24      0   100%

PS C:\Users\Lenovo\Desktop\Sem-5\SE\SE-lab\lab-6\L6_L7_20_food_delivery\Lab 6>
```